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New Interactive Visual Tools and Statistical Methodology for Selecting and Evaluating Non-linear Dimension Reduction Layouts of High-dimensional Data

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Abstract

High-dimensional datasets consist of observations characterized by numerous features, often exhibiting complex geometric properties that can be challenging to visualize. While linear dimension reductions are a reliable method for representing high-dimensional data, they can lead to cluttered visualizations where global patterns obscure local structures and can result in concentrating the data at the center of projections. To overcome these limitations, Non-linear dimension reduction (NLDR) techniques have been developed, applying non-linear transformations to provide clearer, low-dimensional representations of high-dimensional data. However, the effectiveness of NLDR methods can vary based on the choice of techniques and hyper-parameters, leading to diverse representations that can be either accurate or misleading. The main objective of this research is to develop new methods and software tools for understanding, and evaluating NLDR techniques to improve our understanding of high-dimensional data structures.

This research presents four original contributions. The first contribution (Chapter 2) introduces a new method for visualizing how NLDR warps data. This method improves the diagnostics of NLDR techniques. The second contribution (Chapter 3) provides evidence in identification of clusters at various distances when observing NLDR representation and the tour view of high-dimensional data. This finding is based on a human subject experiment that explores both the perception and misperception of NLDR representations. The third contribution (Chapter 4) presents two R packages: `quollr` and `cardinalR`. The `quollr` implements the method introduced in Chapter 2 as an R package. The `cardinalR` is developed to generate high-dimensional clustering data structures, with features such as adding noise dimensions and background noise. Finally, the fourth contribution (Chapter 5) features a Shiny app that offers a user-friendly interface for analysts to obtain the most accurate NLDR representation. Overall, this work advances the field of diagnosing NLDR by improving the visualization of high-dimensional data.

Declaration

I hereby declare that this thesis contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

This thesis includes one original papers published in a peer reviewed journal and four unpublished papers. The core theme of the thesis is to "develop methods and software to understand non-linear dimension reduction methods". The ideas, development and writing up of all the papers in the thesis were the principal responsibility of myself, the student, working within the Department of Econometrics and Business Statistics under the supervision of Professor Dianne Cook, Dr Paul Harrison (MGBP, BDInstitute), Dr Michael Lydeamore, and Dr Thiyanga S. Talagala (University of Sri Jayewardenepura).

The inclusion of co-authors reflects the fact that the work came from active collaboration between researchers and acknowledges input into team-based research. In the case of Chapter 2, Chapter 4, and, Chapter 5, my contribution to the work involved the following:

Chapter	Publication title	Status	Student contribution	Co-authors contribution	Coauthors are Monash students
2		Revised and resubmitted in the Journal of Computational and Graphical Statistics	80% Concept, Analysis, Software, Writing		No

(continued)

Chapter	Publication title	Status	Student contribution	Co-authors contribution	Coauthors are Monash students
4		Submitted in the R Journal	80% Concept, Analysis, Software, Writing		No
4		Submitted in the R Journal	80% Concept, Analysis, Software, Writing		No
5		Submitted in the Oxford Academic (Bioinformatics Advances)	80% Concept, Analysis, Software, Writing		No

Chapters 3 is planned for submission to peer-reviewed journal.

To ensure the clarity and coherence of the written content, artificial intelligence tools were employed to assist in smoothing and refining the language throughout the thesis.

The thesis is written in Australian spelling, except for Chapters 2, 4, and 5, which use American spelling as specified by the publication venue.

I have renumbered sections of submitted papers in order to generate a consistent presentation within the thesis.

Student name: Piyadi Gamage Jayani Lakshika

Student signature:

Date: 13th January 2026

I hereby certify that the above declaration correctly reflects the nature and extent of the student's and co-authors' contributions to this work. In instances where I am not the responsible author I have consulted with the responsible author to agree on the respective contributions of the authors.

Main Supervisor name: Dianne Cook

Main Supervisor signature:

Date: 13th January 2026

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Chapter 1

Introduction

High-dimensional data, where each observation is described by many variables, is increasingly prevalent in modern science from bioinformatics to computer vision. For example, **CITE-seq data** (**?**) simultaneously records RNA expression and protein markers for individual cells, producing rich, complex datasets. To interpret such data, analysts often rely on **dimension reduction** methods to create low-dimensional visualizations that can reveal patterns and structure.

One established approach is **linear projection**, where high-dimensional points are represented as linear combinations of the original features. **Principal Component Analysis (PCA)** (for an overview see Jolliffe (2011)) is the most familiar method, identifying directions of maximum variance. Extending this idea, **tours** (Lee et al. 2021) provide dynamic sequences of linear projections, giving views from multiple angles to help reveal hidden structure. Tour methods are implemented in R packages such as `tourr` (Wickham et al. 2011), `langevitour` (**?**), and `detourr` (Hart and Wang 2022). A key advantage of linear projections is that they preserve the geometric relationships of the original data, they do not introduce distortion. However, linear projections can become cluttered, and global structure may obscure local detail. Furthermore, **piling** (Laa et al. 2022) where points concentrate in the center of projections can mask important variation.

To overcome these limitations, analysts frequently turn to **nonlinear dimension reduction (NLDR)** methods such as tSNE (Maaten and Hinton 2008), UMAP (McInnes et al. 2018), PHATE (Moon et al. 2019), TriMAP (**?**), and PaCMAP (Wang et al. 2021). NLDR applies nonlinear transformations to generate low-dimensional embeddings that aim to preserve local or global data relationships. These methods are designed to **exaggerate structure**, making it easier for analysts to detect patterns that may not be apparent through linear projections.

Yet this strength also introduces a critical risk: **NLDR can hallucinate structure**, creating patterns in the low-dimensional space that do not exist in the high-dimensional data. This issue is strikingly illustrated in **?@fig-NLDR-variety**, where eight different NLDR representations of the same CITE-seq dataset vary dramatically due to differences in method or hyperparameter choices. Such variability raises essential questions: *Which layout can be trusted? Which accurately reflects the high-dimensional data structure?*

Despite the widespread use of NLDR, there is no widely accepted or visually interpretable framework for **diagnosing the reliability of NLDR representations**. Analysts are left to rely on subjective judgment when choosing and interpreting NLDR layouts, without tools to distinguish faithful representations from artifacts. There is also a lack of benchmark clustering datasets with known geometric structure for testing NLDR methods systematically.

In addition to technical gaps, **little is known about how people perceive and misperceive structure in NLDR layouts**. It is unclear how different visualizations influence analysts' conclusions, or how distortions introduced by NLDR affect decision making. Given the critical role of visualization in high-dimensional data analysis, understanding the human perception of NLDR representations is essential.

1.1 Research Objectives

This thesis addresses these challenges through four main objectives:

1. **Develop methods and software to diagnose NLDR techniques**, providing tools to assess whether low-dimensional representations accurately reflect high-dimensional structures.

2. **Conduct a large-scale user study to explore perception and misperception in NLDR representations**, assessing how viewers recognize structure differently in NLDR layouts compared to tour-based views. This cognitive perception experiment will help identify common mistakes made when choosing and reporting structure from NLDR visualizations, and will inform best practices for using these methods in data analysis.

3. **Create benchmark datasets with known geometric properties**, via the `cardinalR` package, to systematically evaluate and validate NLDR methods.

4. **Provide a platform for NLDR evaluation**, combining linked linear projections (such as tours) and NLDR layouts to reveal where distortions arise and how structure is preserved or lost.

1.2 Contribution

By addressing both computational and cognitive aspects of NLDR, this research aims to advance our understanding of when and how NLDR methods can be trusted. The work will provide practical tools, benchmark data, and empirical insights to support the responsible use of dimension reduction techniques in high-dimensional data analysis.

1.3 Thesis Outline

The rest of the thesis is organized as follows:

Chapter 2 introduces an algorithm to assess the NLDR and decide on which, if any, is the most reasonable representation of the structure(s) present in high-dimensional data. We create a model starting with NLDR layout that is then used to display as a wireframe in high dimensions.

Chapter 3 provides empirical evidence on how viewers recognize structure differently when using NLDR layouts versus the tour view, particularly with varying distances between clusters. The findings will help clarify common mistakes made when selecting and reporting structures based on NLDR layouts.

Chapter 4 introduces two R packages developed as part of this research. Section 4.1 presents the implementation of the work is available as an R package, named `quollr`, an acronym to "**qu**estioning how a high-dimensional **o**bject **l**ooks in **l**ow-dimensions using **r**" (**?**). This package also contains a function for performing hexagonal binning using a new approach, for saving langevitour results with a specific projection, and link plots to understand the quirks that occur with different NLDR techniques. Section 4.2 proposes the R package, `cardinalR` (**?**) (**c**ollection of v**ar**ious high-**d**imens**i**o**nal** data structures in **R**), which includes functions to generate high-dimensional clustering data structures, with features such as adding noise dimensions and background noise along with some already generated examples.

Chapter 5 introduces `menuraR` (**m**onitoring **e**mbeddings of **n**onlinear **u**nfoldings for **r**epresentation and **a**nalysis in **R**), a Shiny web application designed to select and evaluate NLDR layouts.

Chapter 6 concludes the thesis, summarises the contribution of the work, and discusses some future plans.

Chapter 2

Visualising how non-linear dimension reduction warps your data

Chapter 3

A user study to explore perception and misperception in NLDR representations

Chapter 4

Software

4.1 quollr: An R Package for Visualising 2-D Models from Non-linear Dimension Reductions in High Dimensional Space

4.2 cardinalR: Generating interesting high-dimensional data structures

Chapter 5

Development of an interactive and dynamic graphics system for NLDR users

Chapter 6

Conclusion and future plans

This thesis presents four key contributions that collectively advance the understanding and evaluation of NLDR methods. The work improves NLDR diagnostics, provides insights into human perception of high-dimensional structures, develops methods for generating clustering data structures, and implements user-friendly software tools that support exploratory analysis and visualization.

6.1 Contributions

The primary contributions of this research are fourfold. First, we introduce a novel method for visualizing how NLDR warps data, thereby improving the diagnostics of NLDR techniques. Second, we conduct a human subject experiment to investigate the perception and misperception of NLDR representations, providing evidence on how clusters at varying distances are identified in comparison to high-dimensional tours. Third, we develop two R packages: `quollr`, which implements the proposed diagnostic method, and `cardinalR`, which generates high-dimensional data structures with enhanced features such as added noise dimensions and background noise. Finally, we create a Shiny application that provides analysts with a user-friendly interface for obtaining the most accurate NLDR representation.

The software outputs of this research have been made publicly available to support transparency and reproducibility. The R package `quollr` has been on CRAN since March 2024 and has received 2950 downloads from the CRAN mirror; its development version is hosted on GitHub at https://github.com/jayanilakshika/quollr. The R package `cardinalR` has been available on CRAN since April 2024 and has received 3151 downloads from the CRAN mirror, with the latest development version at https://github.com/jayanilakshika/cardinalR. A Shiny application for `quollr` is accessible via one of the mirror sites at https://menurar.netlify.app/, with its source code available at https://github.com/JayaniLakshika/menuraR.

The survey web application, **Match-a-roo** (https://ebsmonash.shinyapps.io/Match-a-roo/), was designed and implemented in Shiny to collect participant responses and demographic information. Each subject accessed the survey through the shinyapps.io server.

All materials associated with this thesis are openly available at https://github.com/JayaniLakshika/Monash_PhD_thesis, reflecting the principles of transparency and reproducible research. The thesis itself is written in Quarto (**?**) and published online at https://jayani-lakshika-phd-thesis.netlify.app.

In addition, a number of R packages were essential in the development of this work, including **`tidyverse`** [(?)], **`lmtest`** [(?)], **`kableExtra`** [(?)], **`patchwork`** [(?)], **`glue`** [(?)], **`ggpcp`** [(?)], **`here`** [(?)], and **`knitr`** [(?)]. (Final package list to be confirmed upon completion of writing.)

6.2 Future work

There are several directions that this work can be developed.

6.2.1 Extending our algorithm to NLDR representations beyond 2-D

A potential direction for future work is extending the current algorithm to NLDR results that project into more than two dimensions. While most existing tools including those developed in this thesis focus on 2-D embeddings, exploring projections into higher dimensions like 3-D or 5-D spaces could provide richer structural information in some settings.

Binning into cubes (3-D or higher) could be performed relatively easily and used as the basis for constructing a wireframe representation of the fitted model. The algorithm for convex hull computation in p-dimensions, as described by Barber et al. (1996) and implemented in related software (Laurent (2023)), serves as inspiration for this approach. Alternatively, a simpler method using k-means clustering to obtain centroids in higher-dimensional embeddings might be feasible; however, the challenge would lie in determining how to connect these centroids to form an appropriate wireframe structure.

6.2.2 Scagnostics to evaluate NLDR

One promising direction for future work is the integration of scagnostics ((**?**), (**?**)) as an additional tool for evaluating NLDR results. Scagnostics provide a set of quantitative shape-based metrics (e.g., convexity, skewness, stringiness, clumpiness) that describe the geometric characteristics of scatterplots.

By applying these metrics to 2-*D* scatterplots generated by NLDR methods, we could obtain an objective assessment of how well these methods preserve or distort data structures, particularly in relation to their characteristics (eg: non-linearity).

Moreover, investigating how scagnostic profiles vary with different sample sizes for specific underlying data structures would provide valuable insight into the stability and robustness of NLDR methods. This could help identify which methods are more resilient to changes in sample size and which structures are more prone to distortion under small sample sizes.

6.2.3 Compare prediction approaches

Future work includes evaluating and comparing the prediction capabilities of different NLDR methods. Only some methods such as UMAP provide built-in functionality (Konopka (2023)) to project new high-dimensional observations into an existing low-dimensional embedding. Our approach introduces a general prediction framework that can be applied to any NLDR method. It works by identifying the nearest high-dimensional bin centroid for a new observation and assigning its corresponding 2-*D* centroid from the fitted model.

Having predictions from both the built-in functions (when available) and our centroid-based method allows for direct performance comparisons. This enables a systematic evaluation of how well different approaches preserve structure when projecting new observations into an existing NLDR space.

6.2.4 Interactive diagnostic tool for NLDR evaluation

A promising direction for future work is the development of an interactive tool that enables diagnostic evaluation of NLDR methods, particularly in the context of clustering. Since different NLDR techniques and parameter settings can lead to varied low-dimensional representations and possible misclassifications. It is essential to have tools that help users explore and understand the sources of these discrepancies.

We propose building a Shiny-based interactive application that allows users to upload: 2-*D* and high-dimensional Euclidean distance matrices, NLDR embeddings, and results from spin-and-brush analysis ((**?**), (**?**)).

Spin-and-brush is a dynamic visual method used to explore clustering structures in high-dimensional numerical data. It is especially helpful in identifying the influence of nuisance variables, structural differences among clusters (e.g., shape or variance), and detecting low-dimensional manifolds embedded in higher dimensions. This functionality can be implemented using the `detourr` package ((**?**)), which supports recording and replaying brushing sequences.

The envisioned tool would allow users to select a specific cluster and a data point of interest and inspect how the data point relates to its cluster through interactive 2-*D* and high-dimensional distance visualizations.

The user interface could be organized into two panels. The left panel would display the selected cluster and the specific point within the 2-*D* embedding. The right panel would show a distribution of distances from the selected point to all other points within the same cluster.

Interactive brushing between these panels would help users explore where NLDR methods preserve or distort clustering structure. This tool would not only support more intuitive diagnosis of NLDR performance but could also serve as a foundation for building automated evaluation metrics that align with human interpretation.

6.2.5 Lineup protocols to evaluate NLDR sensitivity and structure preservation

A valuable extension of this work would be to develop lineup-based evaluation protocols ((**?**)) for NLDR methods. Lineups, originally introduced as a statistical inference tool for graphical perception, involve presenting a true data plot randomly embedded among a set of null plots generated under a null model. Observers are asked to identify the plot that appears most different, allowing for an assessment of whether a visual structure stands out beyond what might be expected by chance.

Applied to NLDR, lineups could help evaluate how well a 2-*D* layout preserves the structure of the original high-dimensional data. For example, a lineup could contain one plot of the true NLDR layout and multiple null layouts generated from shuffled or noise-added versions of the data. If participants consistently identify the true layout, it suggests that the NLDR method has effectively preserved meaningful structure.

Lineups could also be extended to study the sensitivity of NLDR methods to hyperparameters. Multiple layouts could be shown, each corresponding to a different hyperparameter setting (e.g., number of neighbors in UMAP or perplexity in tSNE), to evaluate whether small parameter changes lead to perceptually different results. This would allow researchers to quantify the robustness of each method and guide more stable parameter selection.

6.2.6 Visualising experimental designs

The main objective of this tool is to visualise and validate results from experiments. It includes a web application that allows users to easily upload their experiment design data and results data for visualisation. Additionally, we plan to incorporate interactive features such as linked selections and filters. While the tool primarily visualises categorical data, transforming continuous data into intervals can provide a useful way to visualise continuous data as well.

The initial workflow includes importing the experimental design and results, data preprocessing, 2-D static visualization, 2-D interactive visualization, and dynamic visualization. The data preprocessing steps involve mapping the design data and finding missing responses in the results, transforming the data to a wide format to compute the number of responses for each factor level combination (missing combinations are recorded as 0), and converting the data into a long format suitable for visualization. For 2-D static plots, `ggplot2` (**?**) is used to provide a clear view of the distribution of counts across various factor levels. `plotly` (**?**) is used to add interactivity, and hovering over the tiles reveals additional information, enhancing the user's ability to interact with and understand the data. The dynamic visualization will show each vertex as a factor level combination, with jittered points representing the number of responses for each factor combination and edges connected with one level change in a factor. Currently, the `detourr` (**?**) package is used for the implementation.

6.2.7 Investigating perception and misperception in NLDR with additional factors

While our current user study has focused on how the distance between clusters affects human perception of NLDR layouts, there remain several important factors that could further influence perception and misperception. A promising direction for future work is to systematically explore how variations in data characteristics impact a user's ability to correctly interpret dimensionality-reduced representations.

Specifically, we propose extending the perceptual study to consider:

- Background noise: Adding uniformly or normally distributed noise to the data can obscure true structure, and it is important to understand how different NLDR methods handle such interference and how users respond to it visually.
- Number of clusters: As the number of clusters increases, distinguishing them in 2-D may become more challenging, particularly if the separation is subtle or overlaps occur.
- Noise dimensions: Including additional high-dimensional features that contain no signal (i.e., noise variables) can affect NLDR outcomes. We aim to evaluate how this impacts perceived structure.
- Sample size: Varying the number of observations may change both the visual density and the stability of the NLDR projection, influencing the interpretability of patterns.

- Random seed: Since many NLDR methods are stochastic (e.g., tSNE, UMAP), different seeds can lead to different embeddings. It is valuable to understand whether these differences are perceptible to users and how they affect interpretability.

By extending the study to incorporate these data-driven variables, we can build a more comprehensive understanding of when and why human misperception occurs in NLDR layouts, and which methods are more resilient to such distortions. This work will support the development of more robust diagnostics and improve the practical use of NLDR.

6.2.8 Comparative perceptual study of PCA and NLDR Methods

Another valuable direction for future work is to investigate how PCA compares to NLDR methods in terms of human perception and interpretability. PCA is a linear method widely used for its simplicity and mathematical transparency, whereas NLDR methods often involve non-linear transformations and hyper-parameter tuning.

By comparing how users interpret and misinterpret PCA layouts versus NLDR generated layouts, we can gain insights into whether linear techniques are inherently easier to understand or whether they may lead to different types of visual distortions. This work would help clarify when PCA is sufficient for visual analysis and when the added complexity of NLDR is warranted, particularly for exploratory tasks that rely on visual intuition.

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Appendix A

Appendix to “Visualising how non-linear dimension reduction warps your data”

Appendix B

Appendix to “A user study to explore perception and misperception in NLDR representations”

Appendix C

Appendix to “quollr”

Appendix D

Appendix to “cardinalR”

Appendix E

Appendix to “Development of an interactive and dynamic graphics system for NLDR users”